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ELECTRODE ASSEMBLY FOR SECONDARY BATTERY

Abstract

An electrode assembly for a secondary battery having improved safety is provided. According to the embodiments of the present disclosure, a front end and a rear end of a positive electrode plate may be insulated from each other to prevent a short circuit from occurring due to deformation of a front end of the negative electrode plate. A rear end of the negative electrode plate may be insulated to prevent a short circuit between the negative electrode plate and the positive electrode plate. An additional insulating tape may be provided on the rear end of the negative electrode plate to prevent defects due to Z-shaped deformation of the negative electrode plate from occurring.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0029649, filed on Feb. 29, 2024, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

1. FIELD

[0002] Embodiments relate to an electrode assembly for a secondary battery that has improved safety.

2. DESCRIPTION OF THE RELATED ART

[0003] In general, a secondary battery may include an electrode assembly in which a separator is disposed between a positive electrode plate and a negative electrode plate. The separator, the positive electrode plate, and the negative electrode plate may be stacked or wound. A case is provided in which the electrode assembly is accommodated together with an electrolyte, and a cap assembly seals the case.

[0004] If the electrode assembly is wound, a front end of the positive electrode plate near a center of the winding is cut from an active material layer. As the cycle of the secondary battery progresses, the negative electrode plate is repeatedly contracted and expanded to cause deformation of the front end into a Z-shape. A front end of the negative electrode plate may be deformed into a Z shape and be closer to a front end of the positive electrode plate. If the active material at the front end of the positive electrode plate or the exposed base material (non-coated portion) is in contact with the active material of the negative electrode plate that has been deformed into the Z shape, a very strong short circuit may occur. It is desirable to solve this problem.

[0005] The above-described information is disclosed with respect to technology that serves as the background of the present disclosure and is only for improving understanding of the background of the present disclosure and, thus, may include information that does not constitute the related art.

SUMMARY

[0006] Aspects of some embodiments of the present disclosure provide an electrode assembly for a secondary battery having improved safety.

[0007] According to some embodiments, an electrode assembly for a secondary battery includes: a first electrode plate on which a non-coating portion is disposed at a front end that is toward a winding center of the electrode assembly and a non-coating portion is disposed at rear end that is opposite to the front end, the first electrode plate has a surface A and a surface B opposite to the surface A, each of the surfaces A and B is provided with an intermediate non-coating portion that is disposed between the front end and the rear end of the first electrode plate, and a positive electrode tab is provided on the intermediate non-coating portion of the surface B; a second electrode plate on which a non-coating portion is disposed on a front end that is toward the winding center and non-coating portion is disposed at a rear end that is opposite to the front end, the second electrode plate has a surface A and a surface B, each of the surfaces A and B is provided with an intermediate non-coating portion that is disposed between the front end and the rear end of the second electrode plate, and a negative electrode tab is provided on the intermediate non-coating portion of the surface B; a tab cover tape attached to the positive electrode tab; a tab cover tape attached to the negative electrode tab; a first insulating tape attached to the non-coating portions at the front and rear ends of the surface A of the first electrode plate; a second insulating tape attached to the non-coating portions at the front and rear ends of the surface B of the first electrode plate; a third insulating tape attached to the surface A at each of the front and rear ends of the second electrode

plate; and a fourth insulating tape attached to the surface B at each of the front and rear ends of the second electrode plate, wherein the first insulating tape and the second insulating tape are different sizes.

[0008] The surface A of each of the first electrode plate and the second electrode plate may be a surface faces the winding center of the electrode assembly.

[0009] A length of the second insulating tape in a longitudinal direction of the first electrode plate may be longer than a length of the first insulating tape in the longitudinal direction of the first electrode plate.

[0010] The length of the second insulating tape may be greater by about $4\text{ mm}\pm 2\text{ mm}$ than the length of the first insulating tape.

[0011] Each of the first insulating tape and the second insulating tape may be larger than each of the non-coating portions at the front and rear ends of the first electrode plate.

[0012] A difference between a length of each of the first insulating tape and the second insulating tape and a length of each of the non-coating portions at the front and rear ends of the first electrode plate in a longitudinal direction of the first electrode plate may be about 2.5 mm.

[0013] A difference between a width of each of the first insulating tape and the second insulating tape and a width of each of the non-coating portions at the front and rear ends of the first electrode plate in a width direction of the first electrode plate may be about 1.3 mm.

[0014] Each of the third insulating tape and the fourth insulating tape may be spaced apart from a negative electrode active material layer of the second electrode plate in a longitudinal direction of the second electrode plate.

[0015] A distance between each of the third and fourth insulating tapes and the negative electrode active material layer may be about 1 mm or less.

[0016] Each of the third insulating tape and the fourth insulating tape may be larger than each of the non-coating portions at the front and rear ends of the second electrode plate.

[0017] A difference between a width of each of the third insulating tape and the fourth insulating tape and a width of each of the non-coating portions at the front and rear ends of the second electrode plate in a width direction of the second electrode plate may be about $1.2\text{ mm}\pm 1\text{ mm}$.

[0018] In a winding process, a cut end of the first electrode plate is positioned within attachment areas of the first insulating tape and the second insulating tape.

[0019] The cut end of the first electrode plate may be positioned within an overlapping area between the first insulating tape and the second insulating tape.

[0020] The cut end of the first electrode plate may be positioned within attachment areas of the third insulating tape and the fourth insulating tape.

[0021] According to some embodiments, an electrode assembly for a secondary battery includes: a first electrode plate provided with a surface A that faces a winding center of the electrode assembly and a surface B that is opposite to the surface A, wherein each of the surfaces A and B is provided with a non-coating portion that is disposed at a front end that faces the winding center and a non-coating portion that is disposed at a rear end that is opposite to the front end, each of the surfaces is provided with an intermediate non-coating portion provided between the front end and the rear end, and a positive electrode tab is provided on the intermediate non-coating portion of the surface B; a second electrode plate provided with a surface A that faces the winding center and a surface B that is opposite to the surface A, wherein each of the surfaces A and B is provided with a non-coating portion that is disposed on each of a front end that faces the winding center and a rear end that is opposite to the front end, each of the surfaces A and B is provided with an intermediate non-coating portion provided between the front end and the rear end, and a negative electrode tab is provided on the intermediate non-coating portion of the surface B; a tab cover tape attached to the positive electrode tab; a tab cover tape attached to the negative electrode tab; a first insulating tape attached to the non-coating portions of the surface A at the front and rear ends; a second insulating tape attached to the non-coating portions of the surface B at the front and rear ends; a third insulating

tape attached to surface A at each of the front and rear ends; and a fourth insulating tape attached to the surface B at the front and rear ends, wherein sizes of the non-coating portions of the surface B of the first electrode plate are greater than sizes of the non-coating portions of the surface A of the first electrode plate.

[0022] The first insulating tape may be larger than the non-coating portions at the front and rear ends of the surface A of the first electrode plate, and the second insulating tape may be larger than the non-coating portions at the front and rear ends of the surface B of the first electrode plate.

[0023] A length of the second insulating tape in a longitudinal direction of the first electrode plate may be longer by about $4\text{ mm} \pm 2\text{ mm}$ than a length of the first insulating tape.

[0024] A difference between a length of each of the first insulating tape and the second insulating tape and a length of each of the non-coating portions at the front and rear ends of the first electrode plate in a longitudinal direction of the first electrode plate may be about 2.5 mm.

[0025] A difference between a width of each of the first insulating tape and the second insulating tape and a width of each of the non-coating portions at the front and rear ends of the first electrode plate in a width direction of the first electrode plate may be about 1.3 mm.

[0026] Each of the third insulating tape and the fourth insulating tape may be spaced apart from a negative electrode active material layer of the second electrode plate in a longitudinal direction of the second electrode plate.

[0027] A spaced distance between each of the third and fourth insulating tapes and the negative electrode active material layer may be about 1 mm or less.

[0028] Each of the third insulating tape and the fourth insulating tape may be larger than each of the non-coating portions at the front and rear ends of the second electrode plate.

[0029] A difference between a width of each of the third insulating tape and the fourth insulating tape and a width of each of the non-coating portions at the front and rear ends of the second electrode plate in a width direction of the second electrode plate may be about $1.2\text{ mm} \pm 1\text{ mm}$.

[0030] A cut end of the first electrode plate may be positioned within attachment areas of the first insulating tape and the second insulating tape.

[0031] The cut end of the first electrode plate may be positioned within an overlapping area between the first insulating tape and the second insulating tape.

[0032] The cut end of the first electrode plate may be within positioned attachment areas of the third insulating tape and the fourth insulating tape.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] The accompanying drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate exemplary embodiments of the present disclosure and, together with the description, serve to explain principles of the present disclosure. In the drawings:

[0034] FIG. 1 illustrates a schematic cross-sectional view of a state before an electrode assembly is wound according to embodiments;

[0035] FIG. 2 illustrates a schematic cross-sectional view of a portion after the electrode assembly is wound in FIG. 1;

[0036] FIG. 3 illustrates a plan view of both surfaces of a positive electrode plate in FIG. 1;

[0037] FIG. 4 illustrates a plan view of a portion of a surface A of the positive electrode plate in FIG. 3;

[0038] FIG. 5 illustrates a plan view of a portion of a surface B of the positive electrode plate in FIG. 3;

[0039] FIGS. 6 and 7 illustrate schematic cross-sectional views of a state in which the electrode

assembly is wound according to embodiments;

[0040] FIG. 8 illustrates a plan view of both surfaces of a negative electrode plate in FIG. 1; and

[0041] FIG. 9 illustrates a plan view of a portion of the negative electrode plate in FIG. 8.

DETAILED DESCRIPTION

[0042] Embodiments of the present disclosure may be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, the disclosed embodiments are provided so that those skilled in the art will thoroughly understand the present disclosure. Moreover, the disclosed embodiments are provided so that this disclosure will be thorough and complete, and the scope of the present disclosure will be fully conveyed to those skilled in the art.

[0043] In the following drawings, the thickness or size of each layer is exaggerated for convenience and clarity of description, and the same reference numerals in the drawings refer to the same elements. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. In this specification, it will also be understood that if a member A is referred to as being connected to a member B, the member A may be directly connected to the member B or indirectly connected to the member B with a member B therebetween.

[0044] The terms used in this specification are for illustrative purposes of the present disclosure only and should not be construed to limit the meaning or the scope of the present disclosure. As used in this specification, a singular form may, unless definitely indicating a particular case in terms of the context, include a plural form. Also, the expressions “comprise/include” and/or “comprising/including” used in this specification neither define the mentioned shapes, numbers, steps, operations, members, elements, and/or groups of these, nor exclude the presence or addition of one or more other different shapes, numbers, steps, operations, members, elements, and/or groups of these, or addition of these. The term “and/or” used herein includes any and all combinations of one or more of the associated listed items.

[0045] As used herein, terms such as “first,” “second,” etc. are used to describe various members, components, areas, layers, and/or portions. However, it is obvious that the members, components, areas, layers, and/or portions should not be defined by these terms. The terms do not mean a particular order, up and down, or superiority, and are used only for distinguishing one member, component, region, layer, or portion from another member, component, region, layer, or portion. Thus, a first member, component, region, layer, or portion which will be described may also refer to a second member, component, region, layer, or portion, without departing from the teaching of the present disclosure.

[0046] Spatially relative terms, such as “below”, “beneath”, “lower”, “above”, “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. These spatially relative terms are intended for easy comprehension of the present disclosure according to various process states or usage states of the present disclosure, and, thus, the present disclosure is not limited thereto. For example, an element or feature shown in the drawings is turned inside out, the element or feature described as “beneath” or “below” may change into “above” or “upper”. Thus, the term “lower” may encompass the term “upper” or “below”.

[0047] Hereinafter, an electrode assembly for a secondary battery according to embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

[0048] FIG. 1 illustrates a schematic cross-sectional view of a state before an electrode assembly is wound according to embodiments. FIG. 2 illustrates a schematic cross-sectional view of a portion after the electrode assembly is wound in FIG. 1.

[0049] An electrode assembly accommodated in a cylindrical or pouch-type case may be wound into a cylindrical or oval shape. In a winding process, the electrode assembly may be wound in a state in which a positive electrode plate, a negative electrode plate, and a separator between the positive electrode plate and the negative electrode plate may be stacked in a predetermined order.

Generally, the form in which the positive electrode plate, the negative electrode plate, and the separator are completely wound may be referred to as an electrode assembly. The form in which the positive electrode plate, the negative electrode plate, and the separator are stacked before winding the positive electrode plate, the negative electrode plate, and the separator may be referred to as a separate term such as an electrode strip. In some embodiments, this may be only a distinction of terms for convenience, and in the present disclosure, the term “electrode assembly” may include all forms before and after the winding.

[0050] Referring to FIGS. 1 and 2, the electrode assembly according to embodiments may include a first electrode plate **100**, a second electrode plate **300**, and a separator **500**. The first electrode plate **100**, the second electrode plate **300**, and the separator **500** may be wound in a state with the first electrode plate **100**, the second electrode plate **300**, and the separator **500** stacked in a predetermined order. Hereinafter, for convenience, an end facing a winding center may be referred to as a front end (or front end portion), and an end opposite the front end may be referred to as a rear end (or rear end portion). In some embodiments, during the winding, an inner surface facing the winding center may be referred to as a surface A, and an outer surface facing the outside may be referred to as a surface B.

[0051] In some embodiments, the first electrode plate **100** may be a positive electrode plate. A first insulating tape **210** and a second insulating tape **230** may be provided at the front end and a rear end of the first electrode plate **100**. In some embodiments, a tab cover tape (not shown) may be provided on an intermediate non-coating portion **150** on which a positive electrode tab **170** is provided.

[0052] In some embodiments, the second electrode plate **300** may be a negative electrode plate. A third insulating tape **410** and a fourth insulating tape **430** may be provided on an end of the second electrode plate **300** and an area adjacent to the second electrode plate **300**. In some embodiments, a tab cover tape (not shown) may be provided on an intermediate non-coating portion **350** on which a negative electrode tab **370** is provided.

[0053] The separator **500** may be disposed between the first electrode plate **100** and the second electrode plate **300** to prevent short circuiting and to enable movement of lithium ions. The separator may be made of polyethylene, polypropylene, a composite film of polyethylene and polypropylene, but is not limited in this regard. In some embodiments, the separator **500** may be disposed between the first electrode plate **100** and the second electrode plate **300** and on the outside of the first electrode plate **100** (surface that is in contact with the second electrode plate if wound).

[0054] Referring again to FIG. 2, the first insulating tape **210** and the second insulating tape **230** of the first electrode plate **100** may be provided to prevent a short circuit with the second electrode plate **300**. The third insulating tape **410** and the fourth insulating tape **430** of the second electrode plate **300** may block contact between a base material of the first electrode plate **100** and a base material of the second electrode plate **300** and prevent the second electrode plate **300** from being deformed.

[0055] Hereinafter, each of the electrode plate and the insulating tape will be described in more detail.

[0056] FIG. 3 illustrates a plan view of both surfaces of the positive electrode plate in FIG. 1. FIG. 4 illustrates a plan view of a portion of a surface A of the positive electrode plate in FIG. 3. FIG. 5 illustrates a plan view of a portion of a surface B of the positive electrode plate in FIG. 3. FIGS. 6 and 7 illustrate schematic cross-sectional views of a state in which the electrode assembly is wound according to embodiments.

[0057] First, referring to FIGS. 1 to 7, the first electrode plate **100** will be described.

[0058] In some embodiments, the first electrode plate **100** may be a positive electrode plate. The first electrode plate **100** may include a positive electrode base material **110**, which is metal foil such as aluminum or an aluminum alloy. A positive electrode active material layer **120**, such as transition metal oxide, may be provided on both surfaces of the positive electrode base material

110. A compound capable of reversible intercalation and deintercalation of lithium (lithiated intercalation compound) may be used as the positive electrode active material. In some embodiments, one or more types of complex oxides of lithium and a metal selected from cobalt, manganese, nickel, and a combination thereof may be used. The complex oxides may be lithium transition metal complex oxide, and specific examples may include lithium nickel-based oxide, lithium cobalt-based oxide, lithium manganese-based oxide, lithium iron phosphate-based compound, cobalt-free nickel-manganese-based oxide, or a combination thereof.

[0059] Positive electrode non-coating portions **130**, **130'**, **140**, and **140'**, which are areas not provided with the positive electrode active material layer **120**, may be provided between the front end and the rear end of the first electrode plate **100**. For convenience, a non-coating portion disposed between the front end and the rear end may be referred to as an intermediate non-coating portion **150**. A positive electrode tab **170** may be disposed on one surface (surface B) of the intermediate non-coating portion **150'**. The positive electrode tab **170** may be disposed to face one side in a longitudinal direction of the first electrode plate **100**.

[0060] In some embodiments of the first electrode plate **100**, a front end-side positive electrode non-coating portion of a surface A (hereinafter referred to as an A-surface front end positive electrode non-coating portion **130**) may have a length that is less than that of a front end-side positive electrode non-coating portion of the surface B (hereinafter referred to as a B-surface front end positive electrode non-coating portion **130'**). In some embodiments of the first electrode plate **100**, a rear end-side positive electrode non-coating portion of the surface A (hereinafter referred to as an A-surface rear end positive electrode non-coating portion **140**) may have a length that is less than that of a rear end-side positive electrode non-coating portion of the surface B (hereinafter referred to as a B-surface rear end positive electrode non-coating portion **140'**). In some embodiments of the first electrode plate **100**, an intermediate non-coating portion of the surface A (hereinafter, referred to as an A-surface intermediate non-coating portion **150**) may have a length that is less than that of an intermediate non-coating portion of the surface B (hereinafter referred to as a B-surface intermediate non-coating portion **150'**). Here, the 'length' of the non-coating portion may be based on the longitudinal direction (left and right direction in FIG. 3) of the first electrode plate **100**. A first insulating tape **210** may be attached to each of the A surface front end positive electrode non-coating portion **130** and the A surface rear end positive electrode non-coating portion **140** of the first electrode plate **100**. A second insulating tape **230** may be attached to each of the B-surface front end positive electrode non-coating portion **130** and the B-surface rear end positive electrode non-coating portion **140** of the first electrode plate **100**. The insulating tape attached to the front end-side non-coating portion may be referred to as a 'Relamy tape.' The length of each of the A-surface rear end positive electrode non-coating portion **140** and the B-side terminal positive electrode non-coating portion **140'** may be in a range of about 13 mm to about 30 mm.

[0061] The first insulating tape **210** may be made of an insulating material that is attached to the surface A of the first electrode plate **100**. The first insulating tape **210** may be attached to the A-surface front end positive electrode non-coating portion **130** and the A-surface rear end positive electrode non-coating portion **140**. The first insulating tape **210** may have a size greater than that of each of the A-surface front end positive electrode non-coating portion **130** and the A-surface rear end positive electrode non-coating portion **140** to cover all of the A-surface front end positive electrode non-coating portion **130** and the A-surface rear end positive electrode non-coating portion **140**. In some embodiments, a length of the first insulating tape **210** (based on a longitudinal direction of the first electrode plate **100**) may be greater by about 2.5 mm in the left and right direction than each of the A-surface front end positive electrode non-coating portion **130** and the A-surface rear end positive electrode non-coating portion **140**. In some embodiments, a width of the first insulating tape **210** based on a width direction of the first electrode plate **100** may be greater by about 1.3 mm in a vertical direction than each of the A-surface front end positive electrode non-coating portion **130** and the A-surface rear end positive electrode non-coating portion **140**. In some

embodiments, a part of each of the non-coating portions **130**, **140** that is covered by the first insulating tape **210** may be about 2.5 mm in length and about 1.3 mm in width (see FIG. 4). [0062] The second insulating tape **230** may be made of an insulating material that is attached to the surface B of the second electrode plate **300**. The second insulating tape **230** may be attached to the B-surface front end positive electrode non-coating portion **130'** and the B-surface rear end positive electrode non-coating portion **140'**. The second insulating tape **230** may have a size greater than that of each of the B-surface front end positive electrode non-coating portion **130'** and the B-surface rear end positive electrode non-coating portion **140'** to cover all of the B-surface front end positive electrode non-coating portion **130'** and the B-surface rear end positive electrode non-coating portion **140'**. In some embodiments, a length of the second insulating tape **230** based on the longitudinal direction of the first electrode plate **100** may be greater by about 2.5 mm in the left and right direction than each of the B-surface front end positive electrode non-coating portion **130'** and the B-surface rear end positive electrode non-coating portion **140'**. In some embodiments, a width of the second insulating tape **230** based on the width direction of the first electrode plate **100** may be greater by about 1.3 mm in a vertical direction than each of the B-surface front end positive electrode non-coating portion **130'** and the B-surface rear end positive electrode non-coating portion **140'**. In some embodiments, a part of each of the non-coating portions **130'**, **140'** that is covered by the second insulating tape **230** may be about 2.5 mm in length and about 1.3 mm in width (see FIG. 5).

[0063] In some embodiments, because the lengths of the non-coating portions of the surface B are greater than the lengths of the non-coating portions of the surface A, the length of the second insulating tape **230** may also be greater than that of the first insulating tape **210**. In some embodiments, the length of the second insulating tape **230** may be greater by about $4\text{ mm} \pm 2\text{ mm}$ than that of the first insulating tape **210**. The reason in which the second insulating tape **230** is longer than the first insulating tape **210** is because, if the electrode assembly **10** is wound, the insulating tape disposed on the outside has to be longer in order to prevent cracks from occurring in the electrode plate. If the insulating tapes attached to the surfaces A and B of the electrode plate have the same length, or if the length of the insulating tape attached to the surface A that is an inner side is longer than the length of the insulating tape attached to the surface B, there is a risk of cracks occurring in the electrode plate. Referring to FIG. 6, in the case of Comparative Example 1, the lengths of the insulating tapes of the surfaces A and B are the same (e.g., about 18 mm), and ends of the insulating tapes are disposed at the same position. In the case of Comparative Example 2, the insulating tape of the surface A is longer than the insulating tape of the surface B (for example, the insulating tape of the surface A has a length of about 22 mm, and the insulating tape of the surface B has a length of about 18 mm), the ends of the insulating tapes may be disposed at different positions. As the secondary battery is cycled, the negative electrode plate may be repeatedly contracted and expanded, and the insulating tape may serve to prevent the cracks from occurring due to the contraction and expansion. However, in the case of Comparative Example 1, the negative electrode plate the area indicated by an the arrow that is outside the end of the insulating tape may not be protected. In the case of Comparative Example 2, the area indicated by the arrow of the negative electrode plate that is not protected by the insulating tape is greater than that in Comparative Example 1, and there is a high risk of the cracks occurring at the ends of the insulating tape.

[0064] However, if the insulating tape attached to the surface B is longer than the insulating tape attached to the surface A, the second electrode plate **300** may be protected from the contraction and expansion by the insulating tape, and cracks may be prevented from occurring. FIG. 7 illustrates a case in which the insulating tape **230** of the surface B has a length greater than that of the insulating tape **210** of the surface A in Embodiments **1** and **2**. Embodiment **1** is an example in which the A-surface insulating tape **210** is about 18 mm, and the B-surface insulating tape **230** is about 22 mm. Embodiment **2** is an example in which the A-surface insulating tape **210** is about 18 mm, and the

B-surface insulating tape **230** is about 20 mm. In the case of Embodiment **1**, the ends of the insulating tapes **210** and **230** may be disposed at different positions, but the second electrode plate **300** between the two ends may be an area (indicated by the arrow) that is protected by the insulating tape **230**. In the case of Embodiment **2**, the second electrode plate **300** on the area between the ends of the insulating tapes **210** and **230** may be an area (indicated by an arrow) that is protected by the insulating tape **230**. Therefore, compared to Comparative Examples in FIG. **6**, the second electrode plate **300** may be prevented from cracking in the active material layer as the electrode plate contracts and expands.

[0065] The above-described first insulating tape **210** and second insulating tape **230** may be made of the same material. However, for process convenience, the first insulating tape **210** and the second insulating tape **230** may have different colors. It may be advantageous for the insulating tape to be precisely attached by matching a stepped portion (due to a difference in length of the insulating tapes) between the surfaces A and B.

[0066] Referring to FIG. **3**, the second insulating tape **230** may be attached closer to the front end than the first insulating tape **210** in consideration of a distribution margin of the attachment position.

[0067] In some embodiments, if the first electrode plate **100** is cut in the winding process, the cutting may be performed on the area on which the insulating tape is attached. The cutting position may be adjusted and changed within a range of the non-coating portion. However, in embodiments, the cutting position may be disposed within an area in which the first insulating tape **210** and the second insulating tape **230** overlap each other.

[0068] In some embodiments, during the winding process the rear end of the first electrode plate **100** is disposed within an area of the third insulating tape **410** and the fourth insulating tape **430** of the second electrode plate **300**. In some embodiments, the first electrode plate **100** and the size of the non-coating portion of the first electrode plate **100** may be adjusted.

[0069] Hereinafter, the second electrode plate **300** will be described with reference to FIGS. **8** and **9**.

[0070] FIG. **8** illustrates a plan view of both surfaces of the negative electrode plate in FIG. **1**. FIG. **9** illustrates a plan view of a portion of the negative electrode plate in FIG. **8**.

[0071] In some embodiments, the second electrode plate **300** may be a negative electrode plate. The second electrode plate **300** may include a negative electrode base material **310** that is metal foil such as copper, a copper alloy, nickel, or a nickel alloy. A negative electrode active material layer **320**, such as graphite or carbon, is provided on both surfaces of the negative electrode base material **310**. In some embodiments, the negative active material may include a material capable of reversibly intercalating/deintercalating lithium ions, a lithium metal, an alloy of the lithium metal, a material capable of doping and dedoping lithium, or transition metal oxide. The material capable of reversibly intercalating/deintercalating lithium ions may be a carbon-based negative electrode active material and may include, for example, crystalline carbon, amorphous carbon, or a combination thereof. Examples of the crystalline carbon may include graphite such as natural graphite or artificial graphite, and examples of the amorphous carbon may include soft carbon or hard carbon, mesophase pitch carbide, calcined coke, etc. A Si-based negative electrode active material or an Sn-based negative electrode active material may be used as the material capable of doping and dedoping lithium. The Si-based negative electrode active material may be silicon, silicon-carbon composite, SiO_x ($0 < x < 2$), an Si-based alloy, or a combination thereof. The silicon-carbon composite may be a composite of silicon and amorphous carbon. According to some embodiments, the silicon-carbon composite may be in the form of silicon particles and amorphous carbon coated on a surface of each silicon particle. The silicon-carbon composite may further include crystalline carbon. For example, the silicon-carbon composite may include a core containing crystalline carbon and silicon particles and an amorphous carbon coating layer disposed on a surface of the core.

[0072] The negative electrode non-coating portions **330**, **330'**, **340**, and **340'**, which are areas that the negative electrode active material layer **320** is not provided, may be provided between the front end and the rear end of the second electrode plate **300**. For convenience, a non-coating portions disposed between the front end and the rear end may be referred to as intermediate non-coating portions **350** and **350'**. A negative electrode tab **370** may be disposed on one surface (surface B) of an intermediate non-coating portion **350'**. The negative electrode tab **370** may be disposed to face the other side in the longitudinal direction of the second electrode plate **300** (that is, facing a direction opposite to the positive electrode tab).

[0073] In some embodiments of the second electrode plate **300**, a front end-side negative electrode non-coating portion of a surface A (hereinafter referred to as an A-surface front end negative electrode non-coating portion **330**) may have a length that is less than that of a front end-side negative electrode non-coating portion of the surface B (hereinafter referred to as a B-surface front end negative electrode non-coating portion **330'**). In some embodiments of the second electrode plate **300**, a rear end-side negative electrode non-coating portion of the surface A (hereinafter referred to as an A-surface rear end negative electrode non-coating portion **340**) may have a length that is less than that of a rear end-side negative electrode non-coating portion of the surface B (hereinafter referred to as a B-surface rear end negative electrode non-coating portion **340'**). In some embodiments of the second electrode plate **300**, an intermediate non-coating portion of the surface A (hereinafter referred to as an A-surface intermediate non-coating portion **350**) may have a length that is less than or equal to a length of an intermediate non-coating portion of the surface B (hereinafter referred to as a B-surface intermediate non-coating portion **350'**). Here, the 'length' of the non-coating portion may be based on the longitudinal direction (left and right direction in FIG. **8**) of the second electrode plate **300**. The third insulating tape **410** may be attached to each of the A surface front end negative electrode non-coating portion **330** and the A surface rear end negative electrode non-coating portion **340** of the second electrode plate **300**. The fourth insulating tape **430** may be attached to each of the B-surface front end negative electrode non-coating portion **330** and the B-surface rear end negative electrode non-coating portion **340** of the second electrode plate **300**. The insulating tape attached to the front end-side non-coating portion may be referred to as a 'Relamy tape.' In some embodiments, the length of each of the intermediate non-coating portions **350** and **350'** may be less than the length of each of the intermediate non-coating portions **150** and **150'** of the first electrode plate **100**. The length of each of the A-surface rear end negative electrode non-coating portion **140** and the B-surface rear end negative electrode non-coating portion **140'** may be greater than the length of each of the rear end non-coating portions **130** and **130'** of the first electrode plate **100** (see FIG. **1**).

[0074] The third insulating tape **410** may be made of an insulating material that is attached to the surface A of the second electrode plate **300**. The third insulating tape **410** may be attached to the A-surface front end negative electrode non-coating portion **330** and the A-surface rear end negative electrode non-coating portion **340**. The third insulating tape **410** may have a size less than that of each of the A-surface front end negative electrode non-coating portion **330** and the A-surface rear end negative electrode non-coating portion **340**. In more detail with reference to FIG. **9**, the length of the third insulating tape **410** in the longitudinal direction of the second electrode plate **300** may be at least about 8 mm or more. In some embodiments, the length of the third insulating tape **410** is provided so that the position of the end of the rear end-side positive electrode base material does not deviated from the attachment area of the third insulating tape **410** due to winding distribution. In some embodiments, a width of the third insulating tape **410** based on a width direction of the second electrode plate **300** may be greater by about $1.2\text{ mm} \pm 1\text{ mm}$ in a vertical direction than each of the A-surface front end negative electrode non-coating portion **330** and the A-surface rear end negative electrode non-coating portion **340**. In some embodiments, the third insulating tape **410** may be attached to be spaced less than about 1 mm from the negative electrode active material layer **320** in the longitudinal direction of the second electrode plate **300**. This configuration

minimizes thickness distribution due to overlap between the negative electrode active material layer **320** and the third insulating tape **410**. In some embodiments, the insulating tape may be provided at the rear end of the second electrode plate **300** to prevent defects in which the second electrode plate **300** is deformed into the Z-shape due to the contraction and expansion.

[0075] As described above, the insulating tape may be attached to the front end of the positive electrode plate, and the front end of the positive electrode plate may be disposed within the attachment area of the insulating tape of the front end of the negative electrode plate to prevent a short circuit between the positive and negative electrodes from occurring even through the negative electrode plate is formed into the Z-shape.

[0076] According to the embodiments of the present disclosure, the front end and the rear end of the positive electrode plate may be insulated from each other to prevent a short circuit from occurring due to the Z-shaped deformation of the front end of the negative electrode plate.

[0077] In addition, the rear end of the negative electrode plate may be insulated to block a short circuit between the negative electrode plate and the positive electrode plate.

[0078] The above-mentioned embodiments are merely exemplary, and, thus, the present disclosure is not limited to the foregoing embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present disclosure as defined by the following claims.

Claims

1. An electrode assembly for a secondary battery, the electrode assembly comprising: a first electrode plate on which a non-coating portion is disposed at a front end that is toward a winding center of the electrode assembly and a non-coating portion is disposed at a rear end that is opposite to the front end, the first electrode plate has a surface A and a surface B opposite to the surface A, each of the surfaces A and B is provided with an intermediate non-coating portion that is disposed between the front end and the rear end of the first electrode plate, and a positive electrode tab is provided on the intermediate non-coating portion of the surface B; a second electrode plate on which a non-coating portion is disposed at a front end that is toward the winding center and a non-coating portion is disposed at a rear end that is opposite to the front end, the second electrode plate has a surface A and a surface B, each of the surfaces A and B is provided with an intermediate non-coating portion that is disposed between the front end and the rear end of the second electrode plate, and a negative electrode tab is provided on the intermediate non-coating portion of the surface B; a tab cover tape attached to the positive electrode tab; a tab cover tape attached to the negative electrode tab; a first insulating tape attached to the non-coating portions of the surface A at the front and rear ends of the first electrode plate; a second insulating tape attached to the non-coating portions of the surface B at the front and rear ends of the first electrode plate; a third insulating tape attached to the surface A at each of the front and rear ends of the second electrode plate; and a fourth insulating tape attached to the surface B at each of the front and rear ends of the second electrode plate, wherein the first insulating tape and the second insulating tape are different sizes.
2. The electrode assembly as claimed in claim 1, wherein the surface A of each of the first electrode plate and the second electrode plate faces the winding center of the electrode assembly.
3. The electrode assembly as claimed in claim 2, wherein a length of the second insulating tape in a longitudinal direction of the first electrode plate is longer than a length of the first insulating tape in the longitudinal direction of the first electrode plate.
4. The electrode assembly as claimed in claim 3, wherein the length of the second insulating tape is greater by about $4\text{ mm} \pm 2\text{ mm}$ than the length of the first insulating tape.
5. The electrode assembly as claimed in claim 2, wherein each of the first insulating tape and the second insulating tape is larger than each of the non-coating portions at the front and rear ends of

the first electrode plate.

6. The electrode assembly as claimed in claim 5, wherein a difference between a length of each of the first insulating tape and the second insulating tape and a length of each of the non-coating portions at the front and rear ends of the first electrode plate in a longitudinal direction of the first electrode plate is about 2.5 mm.

7. The electrode assembly as claimed in claim 5, wherein a difference between a width of each of the first insulating tape and the second insulating tape and a width of each of the non-coating portions at the front and rear ends of the first electrode plate in a width direction of the first electrode plate is about 1.3 mm.

8. The electrode assembly as claimed in claim 2, wherein each of the third insulating tape and the fourth insulating tape is spaced apart from a negative electrode active material layer of the second electrode plate in a longitudinal direction of the second electrode plate.

9. The electrode assembly as claimed in claim 8, wherein a distance between each of the third and fourth insulating tapes and the negative electrode active material layer is about 1 mm or less.

10. The electrode assembly as claimed in claim 8, wherein each of the third insulating tape and the fourth insulating tape is larger than each of the non-coating portions at the front and rear ends of the second electrode plate.

11. The electrode assembly as claimed in claim 10, wherein a difference between a width of each of the third insulating tape and the fourth insulating tape and a width of each of the non-coating portions at the front and rear ends of the second electrode plate in a width direction of the second electrode plate is about $1.2\text{ mm} \pm 1\text{ mm}$.

12. The electrode assembly as claimed in claim 2, wherein a cut end of the first electrode plate is positioned within attachment areas of the first insulating tape and the second insulating tape.

13. The electrode assembly as claimed in claim 12, wherein the cut end of the first electrode plate is positioned within an overlapping area between the first insulating tape and the second insulating tape.

14. The electrode assembly as claimed in claim 13, wherein the cut end of the first electrode plate is within positioned attachment areas of the third insulating tape and the fourth insulating tape.

15. An electrode assembly for a secondary battery, the electrode assembly comprising: a first electrode plate provided with a surface A that faces a winding center of the electrode assembly and a surface B that is opposite to the surface A, wherein each of the surfaces A and B is provided with a non-coating portion that is disposed at a front end that faces the winding center and a non-coating portion that is disposed at a rear end that is opposite to the front end, each of the surfaces A and B is provided with an intermediate non-coating portion provided between the front end and the rear end, and a positive electrode tab is provided on the intermediate non-coating portion of the surface B; a second electrode plate provided with a surface A that faces the winding center and a surface B that is opposite to the surface A, wherein each of the surfaces A and B is provided with a non-coating portion that is disposed at a front end that faces the winding center and a non-coating portion that is disposed at a rear end that is opposite to the front end, each of the surfaces A and B is provided with an intermediate non-coating portion provided between the front end and the rear end, and a negative electrode tab is provided on the intermediate non-coating portion of the surface B; a tab cover tape attached to the positive electrode tab; a tab cover tape attached to the negative electrode tab; a first insulating tape attached to the non-coating portions of the surface A at the front and rear ends; a second insulating tape attached to the non-coating portions of the surface B at the front and rear ends; a third insulating tape attached to the surface A at each of the front and rear ends; and a fourth insulating tape attached to the surface B at each of the front and rear ends, wherein sizes of the non-coating portions of the surface B of the first electrode plate are greater than sizes of the non-coating portions of the surface A of the first electrode plate.

16. The electrode assembly as claimed in claim 15, wherein the first insulating tape is larger than the non-coating portions at the front and rear ends of the surface A of the first electrode plate, and

wherein the second insulating tape is larger than the non-coating portions at the front and rear ends of the surface B of the first electrode plate.

17. The electrode assembly as claimed in claim 15, wherein a length of the second insulating tape in a longitudinal direction of the first electrode plate is longer by about $4\text{ mm}\pm 2\text{ mm}$ than a length of the first insulating tape.

18. The electrode assembly as claimed in claim 15, wherein a difference between a length of each of the first insulating tape and the second insulating tape and a length of each of the non-coating portions at the front and rear ends of the first electrode plate in a longitudinal direction of the first electrode plate is about 2.5 mm.

19. The electrode assembly as claimed in claim 18, wherein a difference between a width of each of the first insulating tape and the second insulating tape and a width of each of the non-coating portions at the front and rear ends of the first electrode plate in a width direction of the first electrode plate is about 1.3 mm.

20. The electrode assembly as claimed in claim 15, wherein each of the third insulating tape and the fourth insulating tape is spaced apart from a negative electrode active material layer of the second electrode plate in a longitudinal direction of the second electrode plate.

21. The electrode assembly as claimed in claim 20, wherein a distance between each of the third and fourth insulating tapes and the negative electrode active material layer is about 1 mm or less.

22. The electrode assembly as claimed in claim 20, wherein each of the third insulating tape and the fourth insulating tape is larger than each of the non-coating portions at the front and rear ends of the second electrode plate.

23. The electrode assembly as claimed in claim 22, wherein a difference between a width of each of the third insulating tape and the fourth insulating tape and a width of each of the non-coating portions at the front and rear ends of the second electrode plate in a width direction of the second electrode plate is about $1.2\text{ mm}\pm 1\text{ mm}$.

24. The electrode assembly as claimed in claim 15, wherein a cut end of the first electrode plate is positioned within attachment areas of the first insulating tape and the second insulating tape.

25. The electrode assembly as claimed in claim 24, wherein the cut of the first electrode plate is positioned within an overlapping area between the first insulating tape and the second insulating tape.

26. The electrode assembly as claimed in claim 25, wherein the cut end of the first electrode plate is positioned within attachment areas of the third insulating tape and the fourth insulating tape.
